	B.M.S. COLLEGE OF ENGINEERING, BANGALORE-19 (Autonomous Institute, Affiliated to VTU) **Department Name: CSE**

First INTERNALS

Course Code: 23CS6PCMAL	**Course Title:** Machine Learning	
Semester: 6^{th}	**Maximum Marks: 40**	**Date: 20/2/2026**
Faculty Handling the Course:	Rekha G S, Dr.Kayarvizhy N, Dr. Pallavi G B, Saritha A N	
Instructions: ***Internal choice is provided in Part C.***		

PART-A
Total 5 Marks (No choice)

No.	Question	Marks	CO	PO	BL
1a	Explain any four major challenges in machine learning. ● Insufficient Quantity of Training Data ● Nonrepresentative Training Data ● Poor-Quality Data ● Irrelevant Features ● Overfitting the Training Data ● Underfitting the Training Data Explanation of any four	**5M**	**1**	**1**	**1**

PART-B
Total 15 Marks (No Choice)

No.	Question	Marks	CO	PO	BL
2a	You are designing a spam detection system (Spam / Not Spam). • Choose between Linear Regression and Logistic Regression. **Logistic Regression(since it is classification)** • Justify your choice mathematically. Linear regression assumes: $y=a_0+a_1x$ This implies continuous output We instead model the **conditional probability**: The sigmoid ensures: $\sigma(z) = \frac{1}{1+e^{-z}}$ So outputs are valid probabilities. $0<\sigma(z)<1$ • Explain how the output of the chosen model is interpreted. If we define: • $y=1 \rightarrow$ Spam	**5M**	**2**	**2**	**4**

	• $y=0 \rightarrow$ Not Spam Then the model output p represents: **Probability that the email is Spam** Examples: • $p=0.92 \rightarrow$ 92% probability email is spam • $p=0.50 \rightarrow$ 50% uncertainty • $p=0.10 \rightarrow$ 10% probability spam (likely not spam) The model does **not** directly output 0 or 1. It outputs a probability between 0 and 1. • Discuss how decision threshold affects classification. In logistic regression, the model outputs: $$p=P(y=1 \mid x)$$ To convert this probability into a class label, we choose a **decision threshold** τ(Default Threshold (0.5) Predict $y=1$ if $p \geq \tau$				
2b	A hospital develops a disease prediction model using data collected from only one city. When deployed nationwide, accuracy drops • Identify the type of ML challenge involved. **Non representative data, Overfitting** • Explain why model performance dropped. The model performance dropped because the **training data distribution is different from the real-world data.** • Suggest corrective measures. • **Collect More Representative Data** • **Gather more training data**	**5M**	**2**	**2**	**4**
2c	Consider that the following data which has Actual and Predicted values of the model. Actual values: 5, 10, 15, 20 Predicted values: 6, 9, 14, 25 Compute MAE, MSE, RMSE and Interpret result. Soln: MAE = $\frac{1}{n}\sum_{i=1}^{n} \lvert y_i - \hat{y}_i \rvert$ MSE = $\frac{1}{n}\sum_{i=1}^{n} (y_i - \hat{y}_i)^2$ RMSE = $\sqrt{\frac{1}{n}\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}$ **MAE-2, MSE-7,RMSE- 2.65** Formula-2 marks, values-3 marks	**5M**	**2**	**2**	**3**

PART- C
Total 20 Marks (Choice)

<table>
<tr><th>No.</th><th>Question</th><th>Marks</th><th>CO</th><th>PO</th><th>BL</th></tr>
<tr><td>3a</td><td>

A transportation company wants to study the relationship between distance and Fuel used. The collected data is as follows

<table>
<tr><th>Distance in km (X)</th><th>Fuel Used in Liters (Y)</th></tr>
<tr><td>50</td><td>4</td></tr>
<tr><td>100</td><td>8</td></tr>
<tr><td>150</td><td>10</td></tr>
<tr><td>200</td><td>13</td></tr>
<tr><td>250</td><td>15</td></tr>
</table>

Design a regression analysis on the data and predict the fuel consumption for 300 km

Soln: Simple linear regression(show each step for the calculation)

Y= 1.9+0.054X

Fuel for 300kms

18.1 litres

</td><td>10M</td><td>3</td><td>3</td><td>6</td></tr>
<tr><td colspan="5">OR</td><td></td></tr>
<tr><td>3b</td><td>

A manufacturing unit wants to predict daily production (Y) based on:

- Machine Operating Hours (X_1)
- Number of Workers (X_2)

Data collected:

<table>
<tr><th>Hours (X_1)</th><th>Workers (X_2)</th><th>(Y)</th></tr>
<tr><td>4</td><td>10</td><td>200</td></tr>
<tr><td>6</td><td>12</td><td>280</td></tr>
<tr><td>8</td><td>14</td><td>360</td></tr>
</table>

Design a regression equation for the above given data.

Soln: **Multiple linear Regression (show each step of the calculation)**

$Y=40X_1+40X_2-400$

</td><td>10M</td><td>3</td><td>3</td><td>6</td></tr>
<tr><td colspan="5">OR</td><td></td></tr>
<tr><td>4a</td><td>

A company wants to predict house prices based on Size, Location score and Age. Two models are proposed:

- Model A: Multiple Linear Regression
- Model B: KNN Regression

<table>
<tr><th>House</th><th>Size (sq ft)</th><th>Location Score</th><th>Age (years)</th><th>Price (₹ Lakhs)</th></tr>
<tr><td>H1</td><td>1200</td><td>8</td><td>5</td><td>75</td></tr>
<tr><td>H2</td><td>1500</td><td>6</td><td>10</td><td>85</td></tr>
<tr><td>H3</td><td>1800</td><td>9</td><td>3</td><td>120</td></tr>
<tr><td>H4</td><td>1000</td><td>5</td><td>15</td><td>60</td></tr>
</table>

a) Predict the price using Model A and Model B for the given instance

- Size = 1400 sqft, Location Score = 7 , Age = 8 years

Model A: Multiple linear Regression:

$Y = -25 + 0.05X_1 + 8X_2 - 1.5X_3$

</td><td>10M</td><td>2</td><td>2</td><td>4</td></tr>
</table>

	• For Size = 1400 sqft, Location Score = 7 , Age = 8 years $Y = -25 + 0.05(1400) + 8(7) - 1.5(8)$ $Y = -25 + 70 + 56 - 12$ $Y = 89$ Model B: KNN If K=2 - 80 Lakhs If K=3 - 93.3 lakhs b) Compare both the approaches in terms of interpretability (How easily the model's decisions can be explained), Sensitivity to noise and Scalability. **Interpretability** is **High** in Multiple Linear Regression since it is Model based and **Low** in KNN since it is Instance based. **Sensitivity to Noise** **Multiple Linear Regression** Uses all data points Noise influence is distributed Outliers can affect model **KNN Regression** Uses only nearest neighbors If one neighbor has noisy price → prediction changes drastically Highly affected by local outliers				
OR					
4b	A fintech company wants to classify loan applicants as: **1 → High Risk and 0 → Low Risk.** The company has collected the following dataset: Classify new applicant using the probability-based model and the distance-based model.	**10M**	**2**	**2**	**4**

Applicant	**Income (₹ Lakhs)**	**Debt Ratio**	**Credit Score**	**Risk**
A1	3	0.70	580	1
A2	8	0.30	720	0
A3	5	0.50	650	1
A4	9	0.20	750	0
A5	6	0.40	690	0

new applicant has: Income = 5 Lakhs, Debt Ratio = 0.55, Credit Score = 660

The company is evaluating two different approaches:

1. A **probability-based predictive model**, defined as:

$Z = -5 + 0.6(DebtRatio \times 10) - 0.4(Income) - 0.005(CreditScore)$
Classification rule: Risk = 1 if Probability ≥ 0.5.

Compute Z

DebtRatio×10=0.55×10=5.5

$$Z = -5 + 0.6(5.5) - 0.4(5) - 0.005(660)$$
$$Z = -5 + 3.3 - 2 - 3.3$$
$$Z = -7$$

$$P = \frac{1}{1 + e^{-Z}}$$
$$P = \frac{1}{1 + e^{7}}$$
$$e^{7} \approx 1096$$
$$P \approx \frac{1}{1097}$$
$$P \approx 0.0009$$

0.0009<0.5

Classification **Low Risk**

2. A **distance-based similarity model**, where classification is performed using Euclidean distance with **K = 3** and majority voting.

Applicant	Income (₹ Lakhs)	Debt Ratio	Credit Score	Risk	Distance
A1	3	0.70	580	1	**80**
A2	8	0.30	720	0	**60**
A3	5	0.50	650	1	**10**
A4	9	0.20	750	0	**90**
A5	6	0.40	690	0	**30**

Since K=3

- A3 → Risk 1
- A5 → Risk 0
- A2 → Risk 0

Majority vote:

0, 0, 1 → **Low Risk**

Compare both the approaches in terms of interpretability and Suitability for financial risk analysis

Probability-Based Model
Clear coefficients:
Debt ratio increases risk
Income reduces risk
Credit score reduces risk

KNN Model
No coefficients
Decision based on nearby applicants
Hard to explain feature influence
Less interpretable.

Probability model is more interpretable and more suitable for Financial risk analysis.